ARIGATOBANK : Builds highly scalable platform to support high traffic surges with Google Cloud

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ARIGATOBANK

About ARIGATOBANK

ARIGATOBANK Inc. is a Tokyo-headquartered financial services business with a focus on social causes. With the vision of "zeroing people in need of money," it provides services to create a society where people can support and encourage each other.

Industries: Financial Services & Insurance

Location: Japan

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Engine

Google Cloud Armor (https://cloud.google.com/armor/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud Nat (https://cloud.google.com/nat)

Cloud Run (https://cloud.google.com/run/)

Cloud Spanner (https://cloud.google.com/spanner/)

Firebase (https://firebase.google.com/)

o help.

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Google Cloud (https://cloud.google.com/)

Cloud Nat (https://cloud.google.com/nat)

Google Cloud results

- Handles 2,000+ donation projects on the platform without requiring any new maintenance systems
- Delivers scalability while maintaining latency within business requirements to accommodate

Supp
donation campaigns without downtime or delays, even during large traffic surges

traffic increases
- Enables a small team of engineers to complete the project without compromising work-life balance

ARIGATOBANK Inc. is a Tokyo-headquartered financial services business with a focus on social causes. The business runs a platform called kifutown which connects people who want to donate money to those in need. Rather than operate as a traditional bank that handles cash directly, ARIGATOBANK operates as an intermediary between donors and recipients. The number of donation projects to reach 'deposit' status on ARIGATOBANK's kifutown platform has grown to over 2,000 in March 2022. Its shareholders include Yusaku Maezawa, the founder of popular apparel platform, ZOZO.

kifutown features two categories of users, a supporter, who makes a donation, and an applicant, who receives the donation. A supporter launches a donation project and evaluates applications for funds from hopeful recipients.

Maezawa's high-profile involvement in ARIGATOBANK and kifutown heavily influenced decision-making around the platform's infrastructure, according to Takuya Kawatsu, Chief Technology Officer and Engineering Manager, ARIGATOBANK. "For business development, we wanted to expand awareness by using Maezawa-san's personal connections and his ability to spread information on SNS," says Kawatsu. "We had to consider the magnitude of Maezawa-san's influence. For example, we predicted that a single post on Twitter could cause a sudden surge in traffic. Therefore, when developing a system that forms the basis of our business, how that system would respond to sudden changes in traffic is the most important factor, and an issue to be addressed."

ARIGATOBANK Phone App

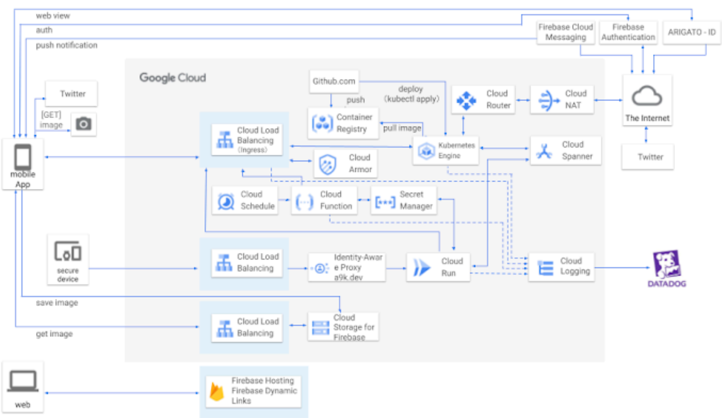
Agility to manage traffic surge challenges with Google Cloud

ARIGATOBANK determined that Google Cloud could overcome these challenges and support new features for kifutown's future growth. kifutown now runs on an architecture centered on Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine/) (GKE), Cloud Spanner (https://cloud.google.com/spanner/), Cloud Load Balancing (https://cloud.google.com/load-balancing/) and Firebase (https://firebase.google.com/), that can scale flexibly and has capacity to support demand. GKE operates as a foundation for microservices and Cloud Spanner as a distributed relational database, while Cloud Load Balancing enables flexible scaling. The architecture also features network security provided by Cloud Armor (https://cloud.google.com/armor/) and networking services delivered through Cloud NAT (https://cloud.google.com/nat) and Cloud Router (https://cloud.google.com/network-connectivity/docs/router/concepts/overview). The business also uses fully managed serverless container platform Cloud Run (https://cloud.google.com/run/) to work efficiently with external SaaS partners.

"We decided in advance to adopt a microservices architecture and at the start of development, there were some areas where we had not yet determined our requirements, so we decided to create the data entity through data modeling and build it as a service," shares Kawatsu, explaining the business decision behind the GKE and Cloud Spanner configuration.

The organization wanted to maintain development speeds and quality even while undertaking progressive development. This meant clarifying the responsibilities for each entity and then applying effort to operating the application server. Takuya chose Kubernetes due largely to the low cost involved, while Cloud Spanner's powerful transaction engine made it the best database choice for kifutown.

Engineering Department representative Tomoki Togashi, charged with developing the service, was deeply impressed with Cloud Spanner and its performance in meeting the organization's requirements. "When I used Cloud Spanner, I realized that the client library was very well done," he says. "The database system also enabled easy management of complex tasks such as transaction conflict detection and retry, while scaling can easily be managed by adjusting settings on a screen."



ARIGATOBANK Architecture

Scaling automatically while maintaining latency

Takahito Yamatoya, in charge of infrastructure development and operation, also emphasizes the importance of flexibility and scalability to the development of kifutown. "With traditional relational databases, there is a limit to scaling up no matter how hard you try, and they take time to respond," says Yamatoya. "With Cloud Spanner, if you increase the number of nodes, it will scale automatically, so you can maintain latency no matter how much traffic you get. This flexibility was essential for kifutown."

In addition, Cloud Spanner enabled the business to use a relational database 'as is', which proved a great advantage in securing experienced engineers. ARIGATOBANK completed the initial development of kifutown in about two and a half months with just four engineers each for back end and front end tasks.

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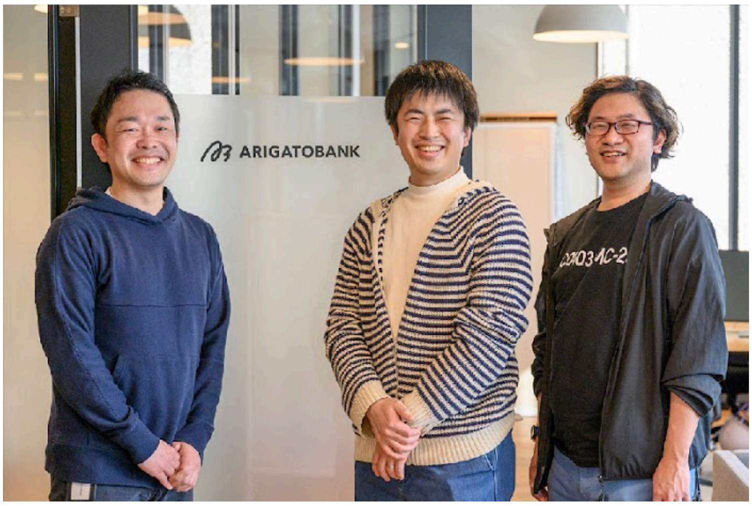
—Takahito Yamatoya, Infrastructure Development and Operation, Engineering Department, ARIGATOBANK Inc.

Rapid development with a small team

According to Togashi and Yamatoya, Google Cloud's GKE and Cloud Spanner, made high-speed development with a lean team possible. "By letting Cloud Spanner handle the processing around transactions, I was able to concentrate on implementing business logic as a developer," says Togashi. "And thanks to GKE, I was able to build infrastructure without distractions." Yamatoya adds, "Thanks to Cloud Spanner, we were able to maintain a simple configuration without having to use extra techniques to maintain scalability and latency." This simplicity enabled the development of kifutown without any major problems.

Kawatsu shares that developing systems at startups often places a heavy burden on engineers, but the use of GKE and Cloud Spanner enabled engineers to focus while maintaining a healthy balance between work and personal lives.

The number of users of kifutown has increased steadily since ARIGATOBANK launched the service and as of March 2022, the number of donation projects to reach 'deposit' status had climbed to about 2,000. This growth has not required Yamatoya to implement any special maintenance systems for kifutown and he says he can run the platform cloud infrastructure with minimal effort. "If there is any issue, we get an alert so we can deal with it immediately," he says. "In the event we are subjected to cyber attacks, Cloud Armor can prevent them. When traffic increases due to a big event, we respond by launching a dedicated service."



ARIGATOBANK Team

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—Tomoki Togashi, Engineering Department representative, ARIGATOBANK Inc.

Supporting a campaign from space

The highest-impact event on kifutown was Maezawa's 'Money Giveaway to Everyone from

Space,' a campaign in which the businessman, who was staying at the International Space Station as a civilian astronaut, made the biggest donation through the platform. The moment Maezawa tweeted about the campaign, traffic surged dozens of times more than usual. "We temporarily increased the number of nodes in Cloud Spanner and significantly expanded the pods in GKE to deal with it. As a result, we were able to get through the surge without any major issues," says Yamatoya.

For ARIGATOBANK, kifutown is the first step towards realizing the vision of "zeroing people in need of money," and the business hopes to create an environment in which individuals can easily donate and help each other.

Kawatsu says he plans to continue updating and growing the platform. "In the past, we used banks to mediate money, but in order to create a better experience for users, we have released an electronic money balance function where users can now receive money as a balance in a wallet on the app and use it as-is, for over-the-counter or online payments."

"We are also considering measures to make it easier to apply for donation projects to increase the overall amount of money distributed. We have a track record of rapid development with Google Cloud, so we would like to continue to use its products, services and technologies in the future," he concludes.

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